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People over-optimistically search for favourite images

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Dear Editorial Board at *Journal of Experimental Psychology: Learning, Memory and Cognition*,

Please find our submission of “People over-optimistically search for favourite images” by van de Wouw, McKay & Furl. Below, we briefly detail why we have selected *Journal of Experimental Psychology: Learning, Memory and Cognition* as a suitable venue for reporting this research and attach an abstract below.

We systematically evaluate alternative computational theories to explain how humans approach decision problems that are ubiquitous in the real world. In these “optimal stopping problems”, sequentially presented options are available for choice only at the time they are presented. Though optimal mathematical solutions exist, most work with human behaviour is limited to a simple “toy” version (the “secretary problem”). Here, we advance our knowledge about humans solve “full information problems”. This computationally more challenging optimal stopping problem better resembles many real-world searches through sequences for ideal options, such as online dating, job searches, shopping, etc. We show that the previously proposed “ideal observer” solution, based only on known ground truth task information, falls short of explaining human choices. We therefore built improved computational theories of choice, each of which includes psychologically interpretable parameterisations to explain choice. We fitted and compared these models to ten datasets, including previously un-investigated search domains. Participants relied mainly on a strategy where they were overly optimistic about future option values and therefore kept searching past optimal stopping points. Using a realistic task that resembles real world decision problems like shopping, we demonstrate how our new modelling approach can explain search biases, replicated across many search domains.

Abstract

In many real-world decisions, options appear one at a time, and choosing an option ends the search. These situations are often described as “fiancé problems,” since one cannot reject a potential marriage partner and later return if a better option does not appear. Decision makers in such scenarios face a computational challenge: They must leverage prior knowledge about the distribution of option values (e.g., the expected value and variability of marriage partners) to predict whether continued sampling of new options might result in a higher-valued option than the one currently on offer. In experimental studies, participants often underperform compared to mathematically optimal algorithms, which are designed to use only ground truth information about the decision problem. Specifically, participants sample more options before accepting one than these algorithms do when the decision involves options presented as images (e.g., searching for attractive faces). Using computational modelling of ten datasets spanning five image-based choice domains, we demonstrate that this apparent oversampling involves optimistic predictions about future option values. Interestingly, our new finding harmonises with previous findings that pessimistic predictions explain the shorter sampling rates observed for economic searches for the best price. These suboptimal search strategies could manifest in real-world decisions such as shopping, choosing dating partners or trustworthy job candidates.

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Furl, N., Averbeck, B. B., & McKay, R. T. (2019). Looking for Mr(s) Right: Decision bias can prevent us from finding the most attractive face. *Cognitive Psychology* 111, 1–14. <https://doi.org/10.1016/j.cogpsych.2019.02.002>

van de Wouw, S., McKay, R., Averbeck, B. J., & Furl, N. (2022). Explaining human sampling rates across different decision domains. *Judgment and Decision Making* 17, 487–512. <https://doi.org/10.1017/S1930297500003557>

RUNNING HEAD: sequential decision strategies

People over-optimistically search for favourite images

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Data and code availability: https://github.com/nicholasfurl/Model_fitting_imageTasks

Abstract

In many real-world decisions, options appear one at a time, and choosing an option ends the search. These situations are often described as “fiancé problems,” since one cannot reject a potential marriage partner and later return if a better option does not appear. Decision makers in such scenarios face a computational challenge: They must leverage prior knowledge about the distribution of option values (e.g., the expected value and variability of marriage partners) to predict whether continued sampling of new options might result in a higher-valued option than the one currently on offer. In experimental studies, participants often underperform compared to mathematically optimal algorithms, which are designed to use only ground truth information about the decision problem. Specifically, participants sample more options before accepting one than these algorithms do when the decision involves options presented as images (e.g., searching for attractive faces). Using computational modelling of ten datasets spanning five image-based choice domains, we demonstrate that this apparent oversampling involves optimistic predictions about future option values. Interestingly, our new finding harmonises with previous findings that pessimistic predictions explain the shorter sampling rates observed for economic searches for the best price. These suboptimal search strategies could manifest in real-world decisions such as shopping, choosing dating partners or trustworthy job candidates.

Introduction

In many real-life situations, decision options appear in sequence and must be selected or rejected at the time of encounter. The behavioural sciences have adopted an approach of comparing decisions on these sorts of sequential choice tasks against mathematical optimality benchmarks (here, termed a “ground truth” benchmark), such as in studies of mate choice in behavioural ecology (Castellano et al., 2012; Castellano & Cermelli, 2011) and economic choices in cognitive science (Baumann et al., 2020) and neuroscience (Costa & Averbeck, 2015). This approach provides a performance benchmark by which researchers can identify which experimental contexts produce better or worse decisions and then develop theoretical models to explain when humans are more or less likely to be accurate.

Consider hiring a new employee. The employer can consider prior knowledge about the probability distribution that generates employees (i.e., option values). Then, the employer interviews a known number of job candidates one at a time such that the quality of each candidate is revealed only at the time of interview. After each interview, the employer must decide whether to hire the candidate or reject them and interview a new one. In this example we assume that competitors will quickly hire rejected candidates and so, once rejected, our employer cannot recall candidates later after rejection. The current state-of-the-art ground truth benchmark for such a task (Costa & Averbeck, 2015) is an evolution of the long-standing Gilbert and Mosteller (1966) benchmark. This benchmark uses only the correct ground truth information, including the mean and variance of a normally distributed prior of the potential option values. Then, using this “generating distribution,” the agent can compute the expected reward value of future samples if the agent were to continue sampling. This expected value functions as a varying choice threshold which the current option value on offer must exceed to be chosen.

Recent work studying how humans solve such sequential decision tasks (Furl et al., 2019; van de Wouw et al., 2022) introduced a new image-based version of the paradigm. Participants

consider images such as faces, foods or holiday destinations and attempt to stop their searches at the best image they can along a predefined dimension (e.g., the most attractive face). Compared to the ground truth benchmark, participants in these image-based studies sampled too many options. Because the ground truth benchmark does not adequately explain participants' sampling rates, we cannot consider it a candidate computational theory to fully explain participants' choices under all circumstances. The current paper builds theoretical computational models to test hypothetical sources of this apparent "oversampling" by fitting the models to participant choice behaviour across ten datasets, including data from Furl and colleagues (2019) and van de Wouw and colleagues (2022) plus three new datasets, testing new image-based choice domains.

Our points of departure are theoretical models previously used in van de Wouw and colleagues (2024). While the computations performed by the ground truth benchmark are a reasonable way to achieve an accurate choice, other psychological contributions must be operative to explain systematic deviations from apparent "optimality." To encapsulate this viewpoint, we parameterised the ground truth benchmark, such that the parameters would represent psychological factors that could contribute to choice. For example, in the *cost to sample* model, we fitted a cost to sample parameter (See General Methods), which reflects how much reward participants subjectively associated with the sampling itself (the ground truth was that the experiment neither charged nor rewarded sample-again decisions). If participants embark on long searches because they perceive sampling to be intrinsically rewarding, we can expect negative values of this fitted parameter. Indeed, the facial attractiveness task (Furl et al., 2019) resembles seemingly addictive dating apps like Tinder, where some users may enjoy evaluating potential mates. Alternatively, in the *biased prior* model, the free parameter is a constant added to the mean of the prior generating distribution. Positive values of this parameter lengthen sampling rates, because the model develops increasingly optimistic expectations about future option values. Lastly, we considered a *cut off model*, which is a parameterisation of the ground truth benchmark for a distinct, but closely related sequential decision problem (Ferguson, 1986; Freeman, 1983). This

ground truth benchmark is technically not the best optimality benchmark for our task because, among several reasons, this benchmark assumes participants only know options' relative ranks and that participants are rewarded only by the highest-ranked options. Nevertheless, instead of using it as a performance benchmark, we consider a parameterised version of it as a potential psychological "heuristic." Todd and Miller (1999) claim that the cut off model's more simple computations are compatible with putative human cognitive limitations and, further, it "satisfices" even when its assumptions are relaxed. During a "learning period" that lasts until a specific cut off sequence position (the free parameter), the heuristic refrains from choice. After the cut off, the choice threshold becomes fixed at the best value observed so far. Longer searches would be associated with later cut offs.

In a previous study (van de Wouw et al., 2024), all these models were fit to sequential choice paradigms where participants searched for the best *numerically presented price in an economic domain*. This previous study showed that the biased prior model was best fitting: Participants stopped sampling earlier than would be expected based on the ground truth model because they had pessimistic beliefs about upcoming option values.

In the present study, we formally compare these models using ten datasets where participants chose *the most appealing image*. To anticipate our results: we replicated participants' higher sampling rates relative to the ground truth benchmark in all ten datasets. Moreover, every dataset showed evidence for the biased prior model as the best fitting. In contrast to the pessimism evinced by participants in numeric-based studies (van de Wouw et al., 2024), participants evaluating images are driven to sample more by optimism about the future, compared to the ground truth model.

General Methods

Paradigm for all datasets

For all datasets, we measured participants' preferences for the options in advance of the decision task by using a phase 1 ratings task, where participants encountered the entire empirical distribution of images, each of which might or might not appear later as options in the decision task later. Participants viewed each image and rated it for "attractiveness" more than once and the multiple ratings were averaged to obtain one stable subjective value estimate per image. We find average correlations across ratings of $> .8$ for all our datasets. We used these subjective value estimates in three ways. First, the subjective values of the same options that a corresponding participant received as option values were input to the ground truth benchmark and all our computational models. The second way we used these subjective values was to program the ground truth benchmark and corresponding computational models with the initial mean and variance of the subject value distribution (before any samples were input). As described below, the ground truth benchmark, the biased prior and cost to sample models employ an internal representation of the generating distribution of options, which is used as a prior when predicting upcoming option values. However, the cut off model is rather different: It assumes participants use no prior at all and its predictions are based instead only on experience of options during the pre-cut off learning period. The third way that we used the subjective values was to gauge the overall success of decision making by using each participant's own subjective values to identify the rank of the chosen options, relative to other images in the same sequence. That is, a well-performing model was one that was able to obtain one of a corresponding participant's favourite images.

Next, each participant and their corresponding model engaged with several sequences of eight image options each, which were randomly sampled without replacement from the images shown in phase 1. These two types of agent (participants and models) decided whether to choose

that option or, instead, reject it, rendering the option inaccessible and producing the next option. Upon choice of an option, all remaining options became inaccessible, yet participants still needed to button press in response to grey squares, which replaced the remaining options. The grey squares were designed to dissuade participants from attempting to finish the study early by sampling fewer options. Our published work shows that the presence of these grey squares does not affect participants' sampling rates (van de Wouw et al., 2024). If either a participant or model reached the final option in a sequence, that option automatically became their choice. A post-sequence feedback display showed the chosen image and solicited from the participant a rating of the reward value for this image. This rating activity only functioned to reinforce the participants' feedback about option value and was not analysed.

Our dependent variable of interest was the sampling rate for the participants, the theoretical models and the ground truth benchmark. We also report in the Supplementary Materials how the chosen price on average ranked within each sequence. Human sampling was statistically compared against these computational measures using an omnibus *F*-test with factor Agent (Human, ground truth benchmark and each of the three fitted models), followed by two-tailed Bayesian pairwise *t*-tests using `bf.ttest` in the MATLAB `bayesFactor` toolbox <https://github.com/klabhub/bayesFactor>. These Bayesian tests use the Jeffreys-Zellner-Siow prior, which involves a Cauchy prior on the effect size with a scale factor of $.71 = \sqrt{2}/2$, expecting medium effect size (Rouder et al., 2009).

Facial attractiveness dataset 1

Behavioural performance and demonstration that participants sample more than the ground truth benchmark for this dataset were reported as Study 2 in Furl and colleagues (2019). The 20 participants were instructed that they would encounter face images of individuals interested in dating them and that the participant's goal was to try to choose the most attractive dates possible.

Participants opted to choose dates from either male or female face sets. Participants in phase 1 rated 426 face images (Bainbridge et al., 2013) for facial attractiveness. Each image was rated three times on an integer scale from 1 to 9. After a brief distractor task, in which participants solved difficult arithmetic problems, they engaged with 28 decision sequences. For each participant, the paradigm generated a fresh random selection (without replacement) of sequence options from phase 1 for every sequence. The facial attractiveness dataset 1 paradigm was programmed in MATLAB R2020b (2010) using the COGENT 2000 toolbox <http://www.vislab.ucl.ac.uk/cogent.php> and data were collected in person in a laboratory setting.

Matchmaker dataset

Analysis of the matchmaker dataset has not been previously published. The paradigm was originally designed to test whether oversampling would replicate when facial attractiveness decisions were divorced from personal mate choice, where participants choose dates for themselves. To test whether participants change strategy when making mate choice decisions for others, we asked participants to imagine they were professional matchmakers and to choose the most attractive faces they could on behalf of a (fictitious) client, who has employed them to select suitable dates. Methods were identical to facial attractiveness dataset 1 aside from these instructions. We enrolled 20 participants (equal gender split), who each chose which gender of face they would like to match for their client. Data were collected in person in a laboratory setting.

Trustworthiness dataset 1

Analysis of trustworthiness dataset 1 has not been previously published. We attempted to replicate oversampling in a paradigm where the option values were still face images but the facial characteristic being maximised by choice was different from attractiveness. We chose a characteristic, trustworthiness, that seemed relevant to real-world contexts, as decisions in contexts like courtrooms can be influenced by facial trustworthiness (Olivola et al., 2014). Participants chose

a face gender for the task, as in the previous datasets. They started the task by rating faces for trustworthiness in phase 1 and then, in the subsequent decision task, trying to choose the most trustworthy-appearing face they could. Methods were identical to face attractiveness dataset 1 and matchmaker dataset, except for these instructions. Most of the 20 participants were female ($N = 18$). Data were collected in person in a laboratory setting.

Trustworthiness dataset 2

Analysis of trustworthiness dataset 2 has not been previously published. This replication study aimed to validate the evidence for oversampling in our previous studies by obtaining an equivalent result of oversampling bias using a larger sample of participants with more data collected per participant. We enrolled from the online Prolific platform (Prolific, 2014) 64 participants between the ages of 18 and 35 years old from the majority English-speaking countries United Kingdom, Ireland, United States, Canada, Australia, and New Zealand. The paradigm was designed and hosted on the Gorilla Experiment Builder platform (Anwyl-Irvine et al., 2020). Participants in phase 1 chose whether they wanted to rate male or female faces and then rated the same images used in our previous studies for trustworthiness. Three images of all the identities were arranged into a random sequence and each image was rated using a 100-point sliding scale, where the slider position was initially hidden and only appeared when the participant made a selection via mouse click. The optimal stopping task immediately followed phase 1, without any intervening distractor task. In the optimal stopping task, participants attempted to choose the most trustworthy face possible. Options were randomly sampled without replacement from the face identities rated in phase 1. Each of the 40 sequences used the same fixed order of options for every participant, although these fixed sequences were presented in a different random order for every participant. Options that remained unsampled in each sequence after choice were replaced by grey squares, which the participant needed to page through via button press. Upon sequence completion, a feedback screen was displayed, as already described for the datasets above.

Facial attractiveness dataset 2, foods dataset 1 and vacations dataset 1

These three datasets were previously analysed in van de Wouw and colleagues (2022), which showed for all datasets that participants sampled more than the ground truth benchmark did. They shared identical methods with trustworthiness dataset 2, apart from the following exceptions. Seventy-five participants per dataset were recruited from Prolific. Participants rated 90 images in phase 1 for “attractiveness”, each twice. They engaged with five sequences during the sequence decision task. In facial attractiveness dataset 2, participants were instructed to choose the most attractive face that they could. In foods dataset 1, stimuli included photographs of prepared meals, (Blechert et al., 2019). In vacations dataset 1, stimuli included images of popular vacation sites (e.g., the Eiffel Tower) taken from a royalty-free image database (www.shutterstock.com). For all datasets, participants were instructed to choose as attractive an option as possible.

Facial attractiveness dataset 3, foods dataset 2 and vacations dataset 2

These three datasets were reported in van de Wouw and colleagues (2022) and replicated the higher sampling rates for participants compared to the ground truth benchmark. The studies were respectively identical to facial attractiveness dataset 2, foods dataset 1 and vacations dataset 1, aside from the participant sample. This participant sample was collected in-person on campus at Royal Holloway, University of London open days from prospective undergraduate candidates and their families. There were 32 participants in facial attractiveness dataset 3, 28 in foods dataset 2, and 36 in vacations dataset 2.

Ground truth benchmark

The benchmark used here is a discrete-time finite-horizon Markov process (Cardinale et al., 2021; Costa & Averbeck, 2015; Furl et al., 2019; van de Wouw et al., 2022). It computes a utility u for states s at samples t , which is taken as the maximal action value, out of available actions a . u

depends in part on the integral over possible future states, weighted by the probability of transitioning into them from the current state $p_t(j|s_t, a)$. This latter term therefore is computed based on the generating distribution, as described above. The model uses backwards induction to derive utility values for the case where the agent chooses to sample again. See Baumann and colleagues (2020) for an intuitive description of backwards induction.

$$u_t(s_t) = \max_{a \in A_{s_t}} \left\{ r_t(s_t, a) + \int_s p_t(j|s_t, a) u_{t+1}(j) dj \right\}$$

u also depends on $r_t(s_t, a)$ the reward to be obtained for action a in the current state. If the agent chooses to decline and sample another option, then $r_t(s_t, a)$ merely equals C , the cost to sample. For all ten datasets, $C = 0$ in the ground truth benchmark, as we never extrinsically rewarded or penalised participants for sampling. However, if the agent chooses one of the options, then the $r_t(s_t, a)$ term in the above formula instead depends on the function R , which maps each option's relative rank h onto its reward value.

$$r_t(s_t, a = \text{take}) = \sum_{i=1}^N p(\text{rank} = i) * R(i + (h - 1))$$

As participants were instructed to choose the best options they could, the ground truth benchmark's choices were always rewarded according to the value of the chosen option (i.e., the average rating that participant gave to the chosen value).

The ground truth benchmark's generating distribution, before any options are sampled, is assigned a prior mean μ_0 , degrees of freedom κ for that prior mean, a prior variance σ_0^2 , and degrees of freedom ν for that prior variance. μ_0 and σ_0^2 were fixed to the mean and variance of the phase 1 rating distribution. The ground truth benchmark then uses Bayes rule to update μ_0 and σ_0^2 upon experiencing each new option value in each sequence.

Fitted models

A theoretically motivated and psychologically interpretable free parameter was fit for each model. For the cut off model, we fit the cut off value, initialising it to the optimal cut off value for the optimal solution and, during fitting, bounded its possible values to be between two (to allow for a learning period of at least one option) and seven (to allow at least one option beyond the learning period to be available for choice). The optimal cut off (and initial value) is the sequence position $1/e \approx 37\%$ through the sequence, which approximates sequence position 3 for our 8 option sequences. For the cost to sample model, we fitted the free cost to sample parameter, initialised it to its ground truth value 0 and bounded it between -100 and 100. Negative values suppress sampling while positive values extend sampling. To maintain continuity with the more traditional use of “cost to sample” as the name of this model quantity (e.g., Costa & Averbeck, 2015), we will retain our previously-used name “cost to sample model” (van de Wouw et al., 2024), even though “reward to sample” might appear to be more apt a description, as this model would explain long searches in terms of sampling being intrinsically rewarding. For the biased prior model, we fit a constant added to the prior mean μ_o , initialising this constant to its ground truth value of 0 and bounded it between -100 and 100. All models additionally included the Softmax inverse temperature parameter β as a free parameter. During fitting, β had an initial value of 1 and was bounded between 0 and 100. Quality of model fit, in the form of the negative log likelihood, summed over each participants’ choices, was derived using probabilities computed from action values passed through the Softmax function. We used `fminsearchbnd.m` in MATLAB (Mathworks, Natick MA) for fitting. Parameter and model recovery is reported in the Supplementary Materials and Figures S1-S4. We computed two metrics for purposes of model comparison. First, we submitted individual participant Bayesian information criterion (BIC) values to Bayesian t -tests (computing as described above for sampling rate comparisons), which involves computing Bayes factors BF_{10} , to test for mean differences in BIC

between pairs of models. Second, we determined which model had the best BIC for the most participants.

Results

The columns of Figure 1 respectively display Bayesian pairwise t-test results for the datasets where participants search for attractive faces as dates for themselves: facial attractiveness datasets 1, 2 and 3. Additional analyses of these datasets, including ranks of chosen items and model parameter estimates, are shown in Supplementary Figure S5.

Insert Figure 1 here

The analysis of these three datasets reproduced those in Furl and colleagues (2019) and van de Wouw and colleagues (2022) and replicated each other, in the sense that participants sampled more than the ground truth benchmark (Figure 1 top row, black and grey bars and points). Indeed, we found significant main effects of Agent in all three datasets: facial attractiveness dataset 1 $F(4, 76) = 115.82, p < .001, \eta^2 = .86$, facial attractiveness dataset 2 $F(4, 296) = 54.49, p < .001, \eta^2 = .42$ and facial attractiveness dataset 3 $F(4, 124) = 29.35, p < .001, \eta^2 = .46$. Figure 1, top row shows Bayesian pairwise test results. Predictive checks on the sampling rate show that all the fitted models could approximate participants' sampling rates, though the cut off model appears to underestimate participants' sampling slightly.

For average model fit (BIC), the models showed a significant main effect in all three datasets: facial attractiveness dataset 1 $F(2, 38) = 55.23, p < .001, \eta^2 = .74$, facial attractiveness dataset 2 $F(2, 148) = 63.83, p < .001, \eta^2 = .46$ and facial attractiveness dataset 3 $F(2, 62) = 13.80, p < .001, \eta^2 = .31$. Figure 1, centre row shows pairwise Bayesian test results. The biased prior model was superior to the other two models in facial attractiveness datasets 1 and 3, and it shared superiority with the cost to sample model over the cut off heuristic in facial attractiveness dataset 3.

The distribution of frequencies of best-fit participants deviated from chance in all three datasets, all with “large” effect sizes (i.e., Cohen's $w > .5$)(Cohen, 1988): facial attractiveness dataset

1 $\chi^2(2) = 34.3$, $p < .001$, Cohen's $w = 1.31$, facial attractiveness dataset 2 $\chi^2(2) = 44.24$, $p < .001$, Cohen's $w = .77$ and facial attractiveness dataset 3 $\chi^2(2) = 10.19$, $p = .01$, Cohen's $w = .56$. The biased prior model appeared superior in fitting participants for all three datasets.

Figure 2 shows results for the three datasets that involve searches through facial image options but were not searches for an attractive date for oneself: matchmaker, trustworthiness 1 and trustworthiness 2. Supplementary Figure S6 displays additional analysis of these datasets, including ranks of chosen items and model parameter estimates.

Participants in all three datasets sampled more than the ground truth benchmark did (Figure 2, top row, black and grey bars and points). This conclusion is supported statistically by main effects of Agent (contrasting participants, ground truth benchmark, and the three theoretical models) in the matchmaker $F(4, 76) = 43.18$, $p < .001$, $\eta^2 = .69$, trustworthiness 1 $F(4, 68) = 106.06$, $p < .001$, $\eta^2 = .86$ and trustworthiness 2 $F(4, 252) = 7.5$, $p < .001$, $\eta^2 = .11$ datasets and by the patterns of Bayesian pairwise tests shown in the top row of Figure 2. Though main effects of Model (contrasting just the three theoretical models) on averaged BIC score were identified for matchmaker $F(2, 38) = 25.54$, $p < .001$, $\eta^2 = .57$, trustworthiness 1 $F(2, 34) = 36.12$, $p < .001$, $\eta^2 = .68$ and trustworthiness 2 $F(2, 126) = 33.06$, $p < .001$, $\eta^2 = .34$ datasets, pairwise Bayesian tests (as shown in Figure 2, centre row) showed that only the trustworthiness 1 dataset showed one model statistically superior to both the other two models: the biased prior model. Analysis of frequency of best-fit participants resembled that shown in Figure 1. The distribution of frequencies of best-fit participants deviated from chance in all three datasets with large effect sizes: facial attractiveness dataset 1 $\chi^2(2) = 9.1$, $p = .01$, Cohen's $w = .67$, facial attractiveness dataset 2 $\chi^2(2) = 10.33$, $p = .01$, Cohen's $w = .76$ and facial attractiveness dataset 3 $\chi^2(2) = 21.22$, $p < .001$, Cohen's $w = .58$. Also as in Figure 1, the biased prior model shown in Figure 2 best-fit more participants than the other two models in all three datasets. Nevertheless, this result was not as clear cut as in the datasets in Figure 1, as the cost to sample model best fit a larger share of participants compared to Figure 1.

Insert Figure 2 here

Figure 3 shows data for the datasets involving non-face image stimuli: foods and vacation destinations. Supplementary Figure S7 shows additional analysis of these datasets, including ranks of chosen items and model parameter estimates.

The top row reproduces the results from van de Wouw and colleagues (2022), showing that all four datasets replicate robust oversampling. This conclusion is supported by main effects of Agent for foods dataset 1 $F(4, 300) = 55.97, p < .001, \eta^2 = .43$, foods dataset 2 $F(4, 108) = 14.84, p < .001, \eta^2 = .35$ vacations dataset 1 $F(4, 296) = 37.23, p < .001, \eta^2 = .33$ and vacations dataset 2 $F(4, 140) = 34.85, p < .001, \eta^2 = .50$ and by the pairwise Bayesian tests shown in the first row of Figure 3. Model comparison results are less obvious for these non-face image domains than the ones involving faces (Figures 1 and 2). All models approximated participants' sampling rates, even if pairwise comparisons between model and human sampling rates were inconsistent across datasets.

As before, main effects of Model on BIC scores were significant for all datasets, foods dataset 1 $F(2, 150) = 121.54, p < .001, \eta^2 = .62$, foods dataset 2 $F(2, 54) = 45.36, p < .001, \eta^2 = .63$, vacations dataset 1 $F(2, 148) = 41.86, p < .001, \eta^2 = .36$ and vacations dataset 2 $F(2, 70) = 3.30, p = .04, \eta^2 = .09$. Pairwise Bayesian tests on BIC scores showed that vacations dataset 1 showed one model – biased prior - had significantly better model fits on average than other models, while both foods datasets showed only that cut off was inferior to the other two models and vacations dataset 2 produced an ambiguous result.

As in Figures 1 and 2, the analysis of frequencies of best fit participants per model showed results that paralleled, yet were more straightforward and interpretable than, analysis of average BIC scores. Frequencies of best fit participants were significantly distributed nonrandomly across models, with “large” effect sizes, in foods dataset 1: $\chi^2(2) = 28.92, p < .001$, Cohen's $w = .62$; foods dataset 2: $\chi^2(2) = 6.93, p = .03$, Cohen's $w = .5$ and vacations dataset 1: $\chi^2(2) = 19.52, p < .001$, Cohen's $w = .51$. In all three of these datasets, the results resembled those in Figure 2: the biased

prior model best-fit more participants than the other two models, while the cost to sample model best fit a notable fraction of participants. However, for vacations dataset 2, we could not conclude that any model statistically “won” more participants than expected by chance alone $\chi^2(2) = .67, p = .72$, Cohen's $w = .14$, consistent with the ambiguous pairwise test results on average BIC scores.

Insert Figure 3 here

Discussion

We built theoretical computational theories to explain why participants make choices sometimes consistent with a computational “optimality” benchmark (which makes decisions solely based on ground truth information) and sometimes inconsistent with this benchmark. We fit these theoretical models to participant choices from ten datasets using formal model comparisons. The biased prior model received the most model fitting evidence, though the results of some datasets provided clearer evidence than others. The evidence favouring the biased prior model was clearest in paradigms where participants searched face images for the one they would consider the most attractive date for themselves. The biased prior also performed strongly when participants searched faces for qualities other than attractiveness or searched non-face images, though these paradigms lead to notable subsets of participants who were best fit by the cost to sample model. Though the biased prior model appears as the dominant strategy, we are open to the idea that participants use multiple strategies, with the strategy mixture varying by domain. Both cost to sample and biased prior models – which are similar because they are based on a parameterised version of the ground truth benchmark – outperformed the cut off model.

Our technique for parameterising ground truth benchmarks, inspired by proposals from Costa and Averbeck (2015) and Todd and Miller (1999), allows us to explain how participants can be highly “accurate” in the sense of making choices that maximise a corresponding participants’ image preferences. At the same time, this approach also can elucidate the subjective psychological processes that motivate participants to deviate from this notion of accuracy. Indeed, Figures S5-S7

show that, across all datasets, while the ground truth benchmark tends to achieve on average its corresponding participants' second favourite image, the participants themselves (and the fitted models) tend to obtain the third or fourth favourite image on average. Thus, though participants can be highly accurate in the same way that the ground truth benchmark is, an optimistic outlook on the prospects of future options may lead participants to sometimes continue sampling beyond the image that the ground truth model would choose.

This approach differs from previous approaches to modelling sequential decision problems and it comes with its own advantages and disadvantages. One advantage is that we build here “cognitive models” that formalise the psychological processes that lead to choice thresholds at each sequence position and, in turn, participants' choices themselves. Parameterised versions of the Costa and Averbeck (2015) model theorise that participants maintain a belief about the probability distribution of option values, which they are continuously updating as they sample new information. Participants' cognitive processes involve using this distribution to compute the expected value of forthcoming option values, which then becomes the choice threshold – the value of the current option must exceed the expected value of future options to be chosen.

In some recent approaches to modelling similar decision problems, more traditional verbal cognitive theories are first associated with predictions about choice patterns, such as distributions of accept responses over sequence positions. Curve fitting methods can then be used to identify the best response distribution (e.g., Lee and Chong, 2024). In another approach, Baumann and colleagues (2023) suggest that linearly decreasing choice thresholds arise from a heuristic that dictates these thresholds, and they assume this heuristic is manageable for participants to compute consciously. In these cases, the computational model describes the response distributions or the thresholds over sequence positions but does not directly embody the cognitive processes that produce them. In contrast, the approach we wanted to take here was to formally simulate where thresholds come from using computational models of the cognitive mechanisms involved. Indeed,

our modelling framework is amenable to the above analysis techniques. One could test whether the biased prior model or any other model from its family produces similar response distributions or threshold patterns as produced by humans and we suggest this fine grained quantitative analyses of model performance as a fruitful area for future research.

Our models of course have limitations as well. It is not clear from our data so far why participants would have such optimistic expectations of option values or whether there are contexts where adopting such an attitude might be advantageous. Moreover, the backward induction technique the biased prior and cost to sample models use to derive expected values of prospective option values involves a rather complex computation that precludes conscious, voluntary or effortful computation by participants. Nevertheless, there is brain imaging evidence that brain responses in fact track backward induction computations (Costa & Averbeck, 2015; Furl & Averbeck, 2011) while does not seem to currently be any evidence for comparable brain-based correlates of more “heuristic” processes. Moreover, it is not well established that the previously proposed “heuristics” are activities that most participants can in fact compute easily in a conscious or deliberate way within working memory constraints. These computations for example involve computing which cut off can “satisfice” (Todd & Miller, 1999) or computing a negatively sloped line, extrapolating points along it, then using those points as decreasing acceptance thresholds (Baumann et al., 2023). Indeed, these models do not specify a computational basis for computing cut offs or decreasing threshold slopes. The general idea of heuristics however is appealing, and we believe future work should indeed be focused on identifying simpler cognitive process models. Models that might approximate backwards induction without its complexity would be invaluable.

As mentioned, the reasons that participants might use a biased prior strategy are not obvious. One possibility might lie in how participants learn their prior distributions. For example, participants may selectively remember high quality options that they hope to select later while forgetting less interesting options. This selective learning of certain high-quality items could occur

during encounters with stimuli (e.g., faces) in the outside world before the study, or during phase 1 within the study. Another possibility is that exposure to the deluge of media content outside of the study, especially commercial advertising, might positively bias participants' expectations about the attractiveness of images depicting faces, foods or vacation destinations. Though these options sound plausible, any viable theory should also explain why participants appear to have pessimistic expectations when the task involves searches through numeric prices, which convey option values without use of images. Indeed, van de Wouw and colleagues (2024) showed that the biased prior model was also the best fitting in this numeric price-based task, even though participants consistently searched through fewer prices than the ground truth benchmark. Why optimism for some domains and pessimism in others? We recommend that future studies concentrate on examining how participants form expectations about reward from statistical distribution learning, and to test whether optimistic prior expectations about images might arise in contexts more generally than just optimal stopping problems.

Though the tasks we studied here all involved image-based searches, many of the methods details varied drastically from dataset to dataset and yet our results remain remarkably consistent, especially the robust finding of that participants sample more than the ground truth benchmark. Perhaps most importantly, the tasks varied by domain. Long searches appear to be a broader phenomenon that generalises beyond personal mate choice (Furl et al., 2019) and extends to mate choices for others (matchmaker dataset), other judgments about faces (trustworthiness datasets 1 and 2) and beyond faces altogether (foods datasets 1 and 2 and vacations datasets 1 and 2). We suspect that it is the image-based nature of the option value presentation that, at least in part, instigates this oversampling bias.

In addition to the searches involving different domains, the ten datasets also vary in whether they were conducted in the lab, were sampled in-person from mostly students on campus (facial attractiveness dataset 1, matchmaker and trustworthiness dataset 1), from worldwide via an

online participant recruitment platform (trustworthiness dataset 2, facial attractiveness dataset 2, foods dataset 1, vacations dataset 1) or in-person, sampled from university open day attendees in the UK (facial attractiveness dataset 3, foods dataset 2, vacations dataset 2). The datasets also varied in sample size from 20 (facial attractiveness dataset 1, matchmaker dataset and trustworthiness dataset 1) to 75 (Facial attractiveness dataset 2, foods dataset 1 and vacations dataset 1). The number of sequences per participant varied from only five (Facial attractiveness datasets 2 and 3, foods dataset 1 and 2 and vacations dataset 1 and 2) to 40 (trustworthiness dataset 2). There was also no systematic relationship between the best-fitting models and the number of times participants rated potential options in phase 1 nor whether there was an intervening distractor task between phase 1 and the full information decision task.

In sum, our study demonstrates the utility of a cognitive modelling-based approach. We show here that an optimistic computational strategy can explain relatively long searches through images. With increasing use of online and smartphone applications, from ordering a meal online to choosing a date on Tinder, more and more decisions are made by accepting and rejecting sequences of images. By studying these types of searches, providers and users of services involving such searches can better understand and perhaps even improve the performance of searches in real world settings like these.

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Figure 1

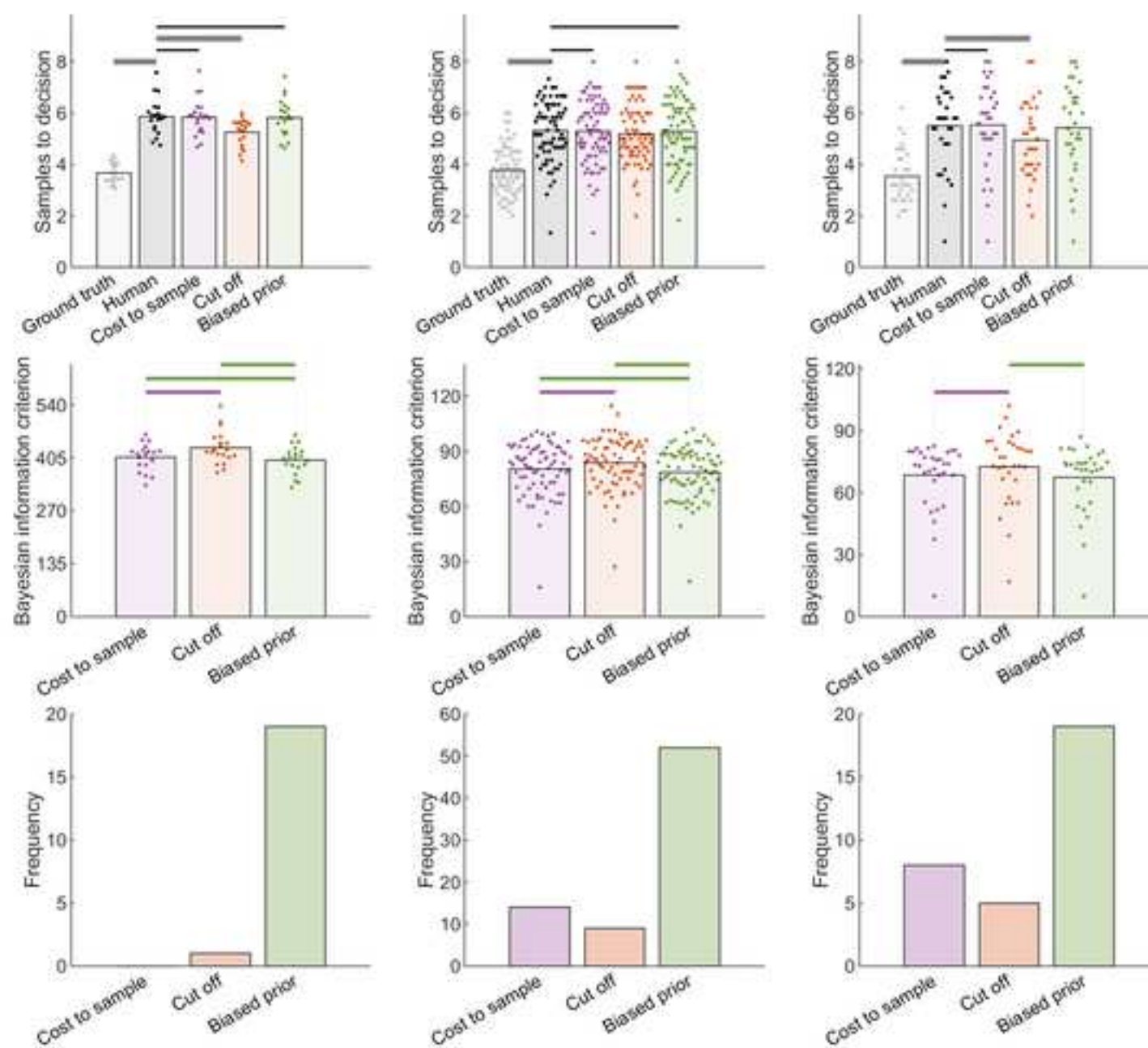


Figure 2

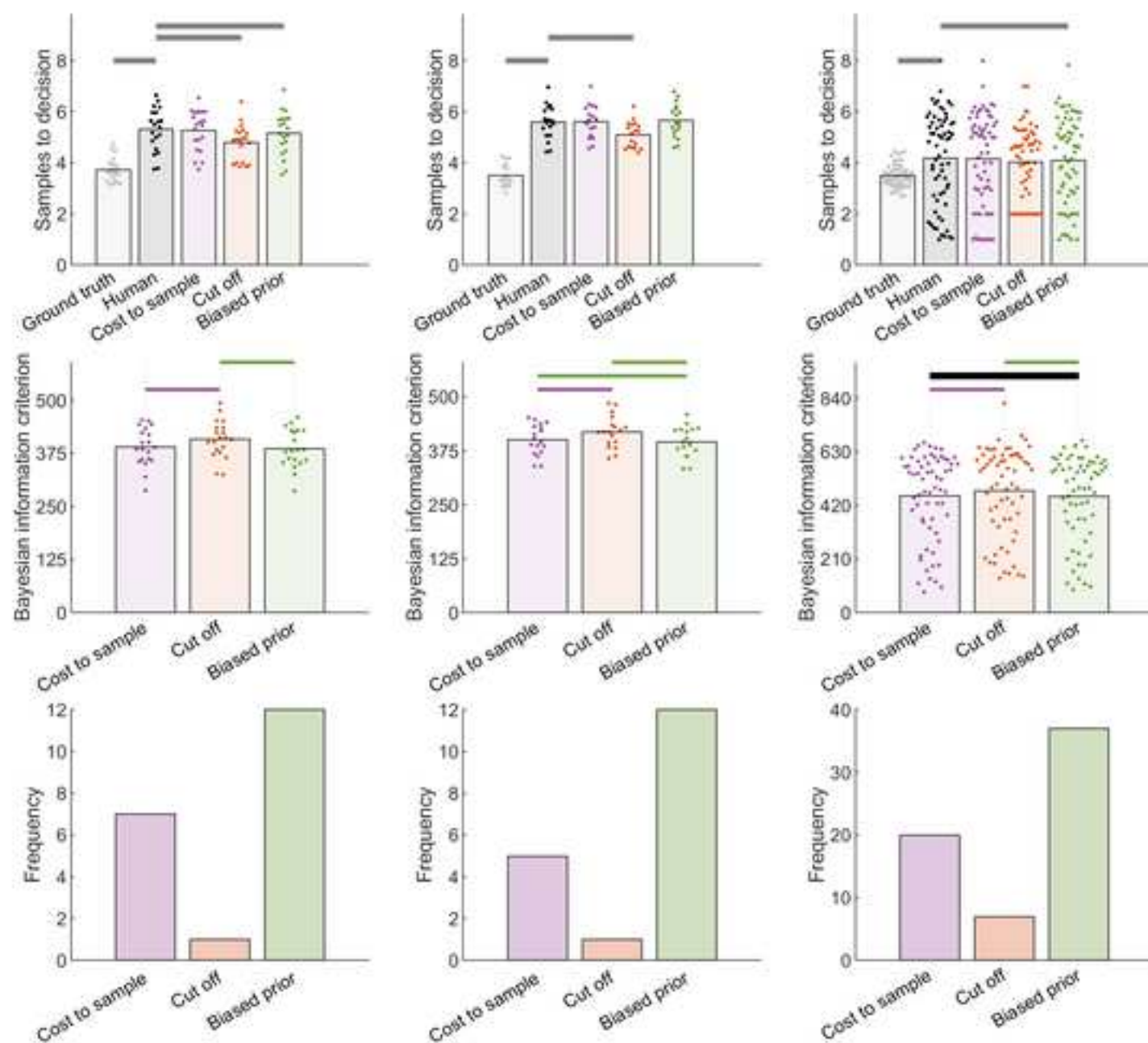
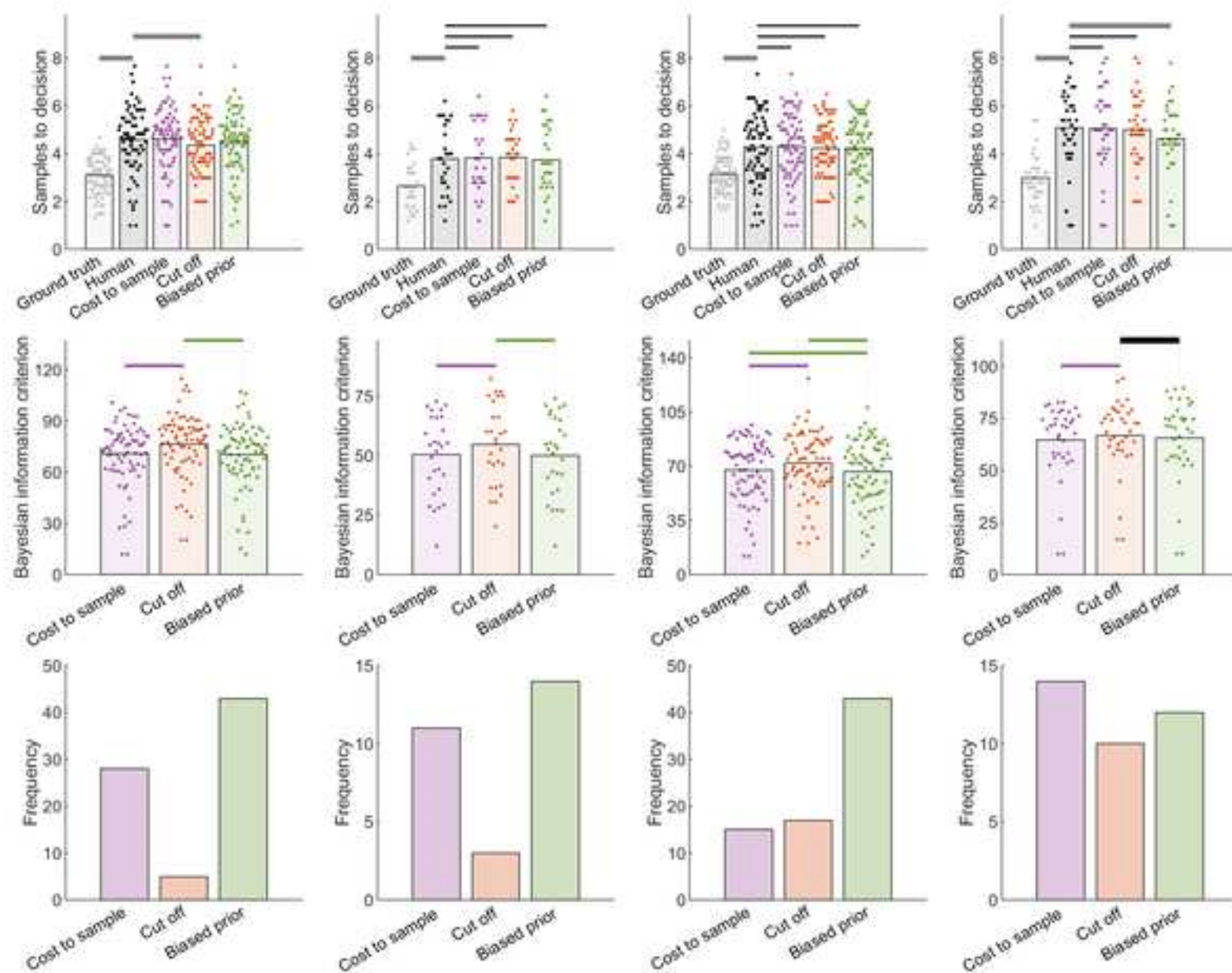
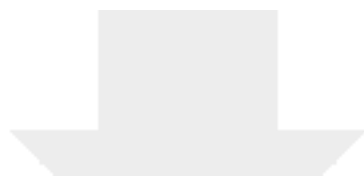


Figure 3





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Supplemental Material

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